

PREPARED FOR AURUMIX

Aurumix AURX: Secondary Market Liquidity Audit

How deep a stablecoin pool AURX needs, what trades will cost in it, and how much of it is gold.

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Executive summary

AURX can trade on a secondary market, and the Mechanism Design Document already says not to seed a thin pool at launch. This audit sizes a pool that is not thin: one where everyday amounts trade at almost exactly the gold price and the largest bar can still exit inside the entry fee. It covers liquidity only, because AURX has no allocations, investors or unlocks to audit.

USD 870k

RECOMMENDED POOL: USD 449K
STABLECOIN PLUS 2.97 KG OF
GOLD

14×

DEPTH OF A $\pm 10\%$
CONCENTRATED POOL AGAINST
A SIMPLE POOL, SAME CAPITAL

USD 12.1m

WHAT A SIMPLE (V2) POOL
WOULD NEED FOR THE SAME
SERVICE

1

A simple pool does not work for AURX

A V2 pool spreads its capital across every price, most of which gold will never reach. At USD 1m it takes only USD 1,765 of selling inside 1%, and a 100 g holder moves its price about 5.7%. Holding the service standard would need USD 12.1m.

2

A concentrated pool does, for under a million

Two thirds of the capital in a $\pm 10\%$ band around the gold price: USD 870k meets the standard. A USD 2,000 trade costs 0.37%, a 100 g exit 0.76%, and a 1 kg exit stays inside the 5% entry fee. A $\pm 5\%$ band does the same with USD 445k.

3

The band needs an owner

Gold leaves a $\pm 5\%$ band within a month in two runs out of five, and a $\pm 10\%$ band in 3%. Once outside, capacity falls 40 to 85 times and the pool shows the discount the design forbids. $\pm 10\%$ needs a quarterly review; $\pm 5\%$ needs a market maker.

4

The pool undercuts the entry fee

A buyer who can reach the pool pays about 0.3 to 0.8% over gold instead of 5%. That is fine if it is a deliberate choice for existing holders and crypto-native buyers. Whether Aurumix may provide the liquidity itself is a question for counsel.

01

What this audit covers

What a secondary market means for a gold-backed token, why only liquidity is audited, and how the pool sits against Aurumix's own doors.

Why this audit

AURX is an open ERC-20 token. One AURX is one gram of allocated gold.

- Any holder can send AURX to anyone else.
- So AURX can trade on a secondary market that Aurumix does not run.
- The Mechanism Design Document (section 4.5) sets the order: a VARA-licensed exchange first, then a broader exchange, then decentralised venues.
- It also sets one rule: **do not seed a thin pool at launch**. A thin pool shows a visible discount to the gold price.

This audit answers three questions:

1. How much does a trade move the price in an AURX/stablecoin pool?
2. How big must the pool be for trades to clear at a fair cost?
3. How much of the pool is stablecoin, and how much is gold?

What is left out

A standard audit also covers allocations, vesting, investor discounts, investor returns and unlock sell pressure. None of these apply to AURX.

- AURX has no allocation table.
- No investor holds discounted tokens.
- Nothing unlocks over time.
- Every AURX is minted when a customer's money buys a gram, and burned when that gram is sold back.

How AURX differs from a launch token

1. THE PRICE IS SET BY GOLD.

- AURX tracks the LBMA gold price per gram (the fix).
- The pool's job is to stay close to that price.

2. BOTH SIDES OF THE POOL COST REAL MONEY.

- Every AURX in the pool is backed by a real gram in the vault.
- A USD 1m pool holds about USD 0.5m of stablecoin and USD 0.5m of gold.
- Aurumix still owns the gold. It is capital tied up in the pool.

3. THE CUSTOMER HAS TWO OTHER DOORS.

- **Buying:** pay the fix plus the entry fee (5% at launch) through the Aurumix app.
- **Selling:** use the buyback. It pays the fix with no fee (VARA Annex 2 III.E.4).
- The pool only wins a trade when it is cheaper than these doors.

Where the pool price will sit

The two doors set a floor and a ceiling.

	PRICE	WHAT HAPPENS
Floor	The fix	Below it, a verified holder buys in the pool and sells back to Aurumix for free. The pool price rises.
Ceiling	The fix plus about 5%	Above it, buying through the app is cheaper. The pool price falls.

Two things follow.

- **Sellers compare the pool with a free buyback.** They use the pool only if slippage is small. So the tightest ceiling in this audit is 0.5%.
- **Buyers get a cheaper way in.** A buyer who can reach the pool pays the fix plus a fraction of a percent, instead of the fix plus 5%.

Method and inputs

The pool maths follows the Tokenomics.net audit engine. Appendix A lists the formulas and the checks.

INPUT	VALUE	SOURCE
AURX price	USD 141.50 (1 gram)	Phase 4 and 5 gold price
Pool fee	0.30% on both designs	Uniswap V2 standard
Slippage ceilings	0.5%, 1%, 2%, 5%	Tighter than the standard 2, 5 and 10%, because of the two doors
Pool sizes tested	USD 100k to 2.5m, both sides combined	No token raise to size from
Trade sizes	USD 1k, 5k, 10k, 20k, 50k	Standard audit grid
Aurumix trade sizes	USD 75 (monthly SIP), USD 2,000 (top SIP ticket), 100 g (USD 14,150), 1 kg (USD 141,500)	Product and bar sizes
V3 layout	Two thirds of capital in a band around gold, one third across all prices	Standard audit layout
V3 bands	±5% and ±10%	Gold moves about 4% in a typical month
Gold volatility	15% a year	Phase 5 simulation

How slippage is measured. A 1% ceiling allows the 0.3% fee plus 0.7% of price movement. The seller's average price is better: a 1 kg sale measured at 4.79% receives 2.56% below gold.

02

The simple pool (Uniswap V2)

The standard constant-product pool: price impact, maximum trade size and the capital it needs.

A V2 pool spreads its capital across every price, from zero upwards.

- It needs no management.
- It never runs out of either asset.
- Most of its capital sits at prices AURX will never reach, such as half or double the gold price.

Price impact

How to read it. Sale size runs along the bottom. Slippage runs up the side. Each line is one pool size. A trade fits a ceiling when its line is below that dashed line.

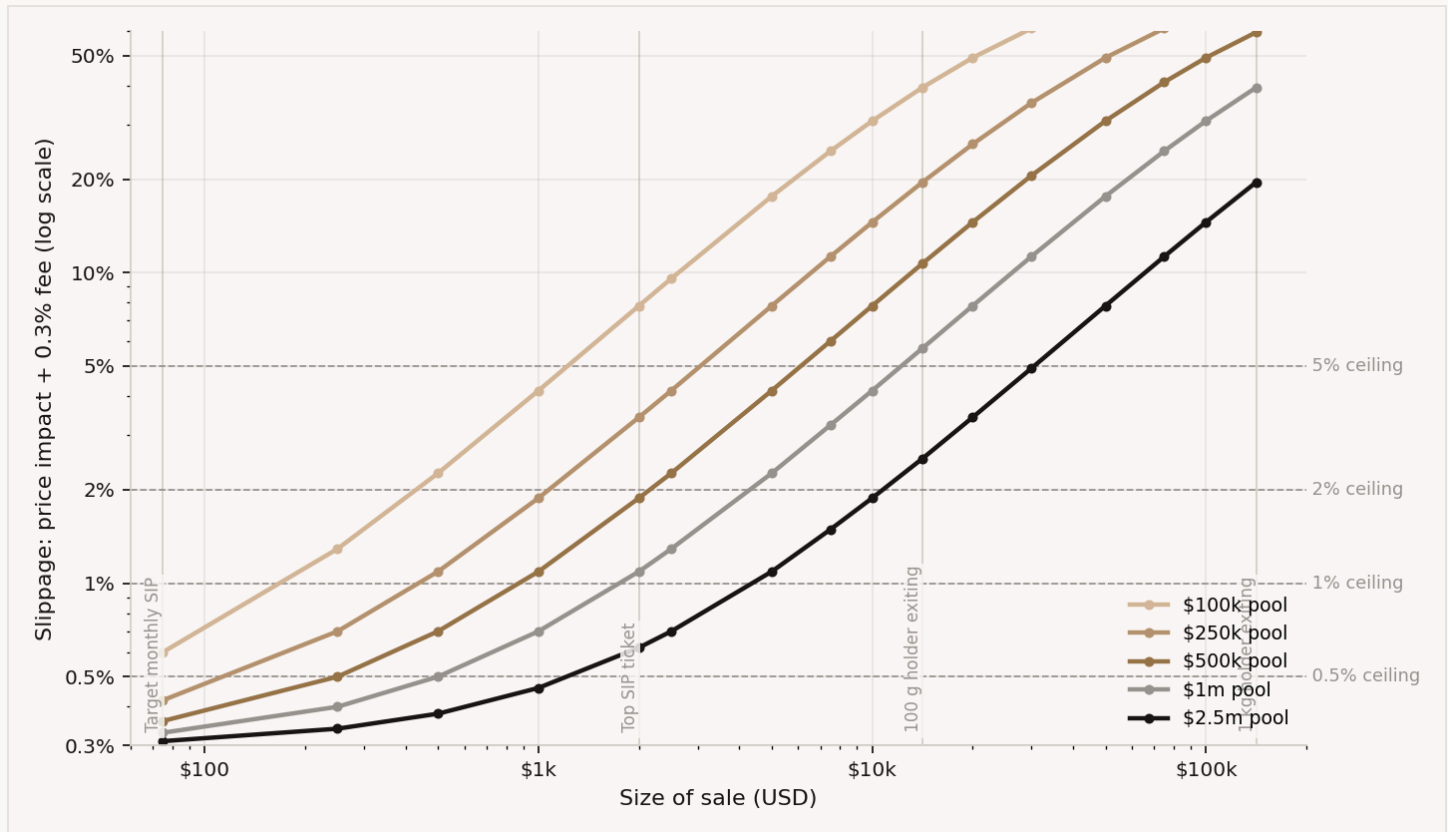


Figure 1. Slippage against sale size in a V2 pool, one line per pool size

WHAT IT SHOWS

- From USD 250k to USD 1m, only the USD 75 SIP payment clears 0.5%.
- A USD 1k trade needs the USD 2.5m pool to clear 0.5%.
- In a USD 1m pool, a USD 2,000 sale costs about 1.1%.
- In the same pool, a 100 g sale costs about 5.7%.
- A 1 kg sale costs about 20% even in the USD 2.5m pool.

Why. Price impact grows with trade size divided by pool size. Double the trade and you need double the pool.

KEY INSIGHTS

- A V2 pool cannot hold AURX near gold for anything larger than a SIP payment.
- The 0.3% fee uses most of a 0.5% ceiling. Only 0.2% is left for price movement.
- Large exits are out of reach at every tested size.

POTENTIAL RISKS

- A thin V2 pool shows AURX below gold. The Mechanism Design Document prohibits this.
- A large seller who will not wait for the fix moves the displayed price. The gold behind each token has not changed.

Maximum trade size

How to read it. For each ceiling, the largest sale the pool can take. One bar per pool size. Each gridline is ten times the one below.

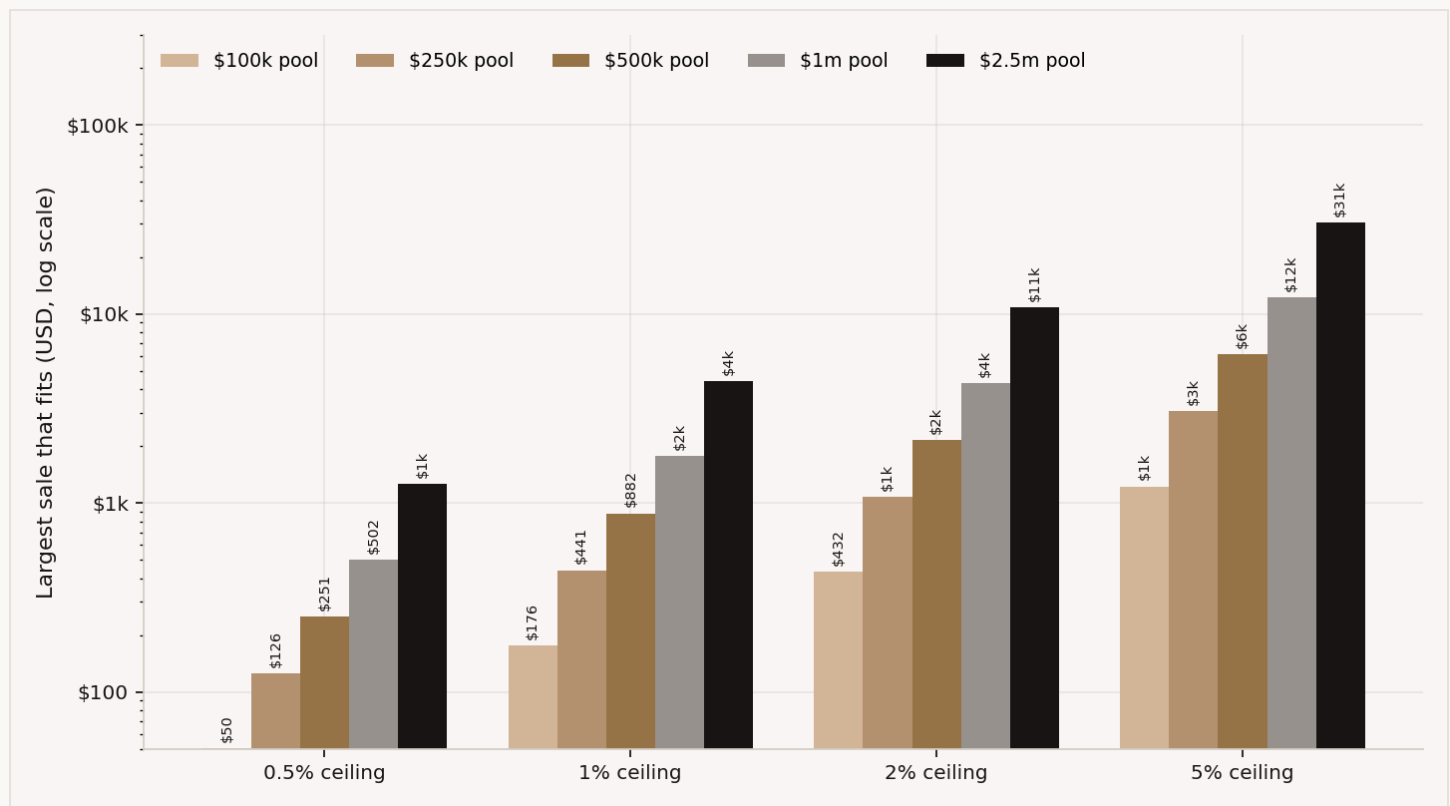


Figure 2. Largest sale inside each ceiling, V2 pool

POOL SIZE	0.5% CEILING	1% CEILING	2% CEILING	5% CEILING
USD 100k	USD 50	USD 176	USD 432	USD 1,222
USD 250k	USD 126	USD 441	USD 1,079	USD 3,054
USD 500k	USD 251	USD 882	USD 2,159	USD 6,109
USD 1m	USD 502	USD 1,765	USD 4,318	USD 12,218
USD 2.5m	USD 1,256	USD 4,411	USD 10,795	USD 30,544

- Double the pool and every cell doubles.
- Buys are slightly smaller than sells: USD 501 and USD 1,752 in the USD 1m pool.

KEY INSIGHTS

- The USD 2.5m pool cannot take one 100 g sale inside 2%. Its limit is USD 10,795, about 76 g.
- The standard audit benchmark is USD 10k to 50k inside 2 to 3%. No V2 pool tested meets it.
- Buy and sell limits are within about 5% of each other.

Liquidity required

How to read it. The pool size needed for each trade to clear each ceiling. One line per trade size. Where a buy needs more than a sale, the larger figure is shown.

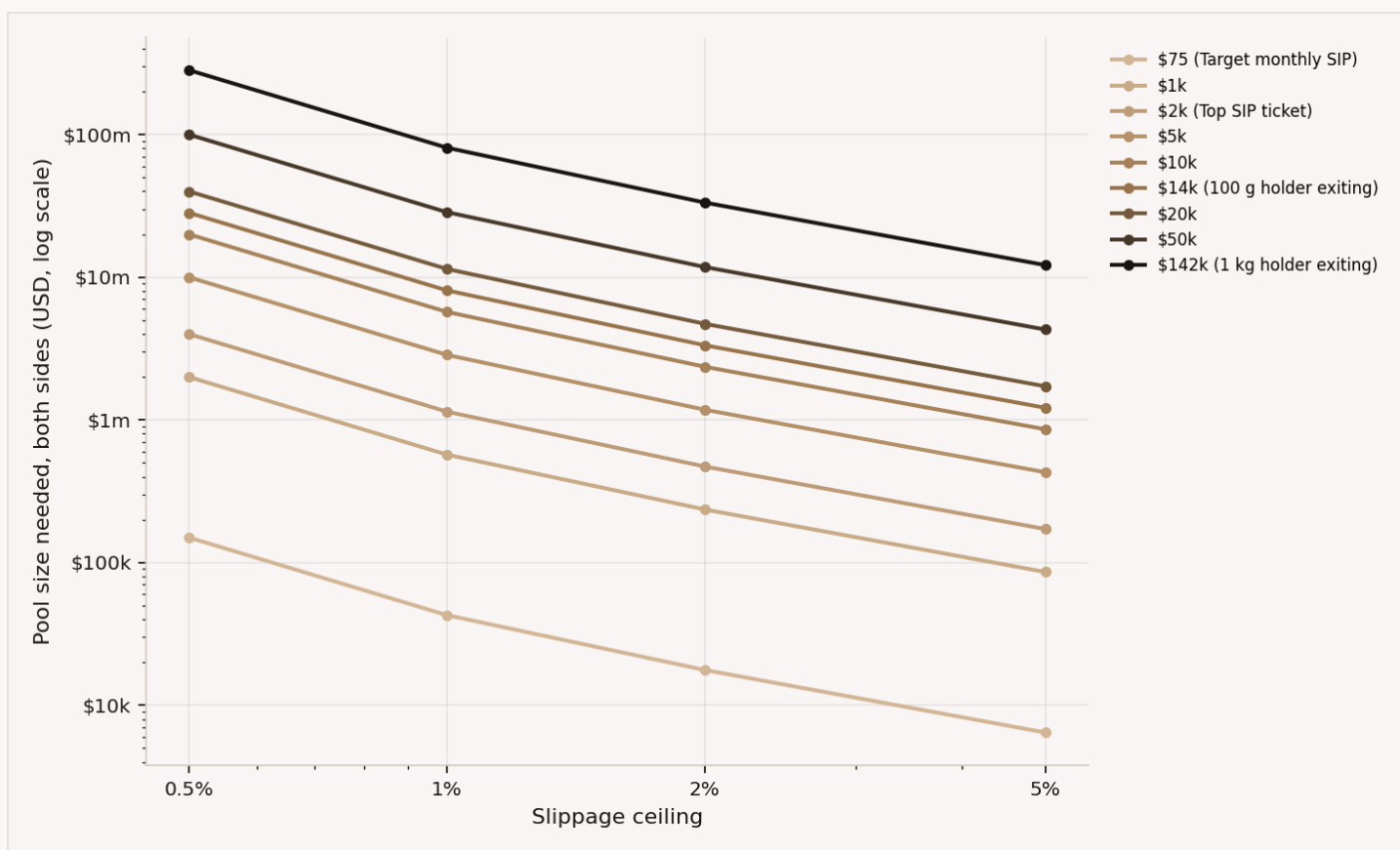


Figure 3. Pool size needed against slippage ceiling, V2, one line per trade size

TRADE	0.5%	1%	2%	5%
USD 75 (target SIP)	USD 150k	USD 43k	USD 18k	USD 6k
USD 2,000 (top SIP ticket)	USD 3.99m	USD 1.14m	USD 471k	USD 172k
USD 10,000	USD 19.9m	USD 5.71m	USD 2.36m	USD 858k
100 g (USD 14,150)	USD 28.2m	USD 8.08m	USD 3.33m	USD 1.21m
1 kg (USD 141,500)	USD 282m	USD 80.8m	USD 33.3m	USD 12.1m

The grid below gives the tightest ceiling each trade clears at each pool size. Darker is better.

Size of sale	\$75 Target monthly SIP -	≤1%	≤0.5%	≤0.5%	≤0.5%	≤0.5%
	\$1k -	≤5%	≤2%	≤2%	≤1%	≤0.5%
	\$2k Top SIP ticket -	>5%	≤5%	≤2%	≤2%	≤1%
	\$5k -	>5%	>5%	≤5%	≤5%	≤2%
	\$10k -	>5%	>5%	>5%	≤5%	≤2%
	\$14k 100 g holder exiting -	>5%	>5%	>5%	>5%	≤5%
	\$20k -	>5%	>5%	>5%	>5%	≤5%
	\$50k -	>5%	>5%	>5%	>5%	>5%
	\$142k 1 kg holder exiting -	>5%	>5%	>5%	>5%	>5%
		\$100k pool	\$250k pool	\$500k pool	\$1m pool	\$2.5m pool
		Pool size, both sides				

Figure 4. Tightest ceiling each trade clears, by pool size, V2

- Below USD 2.5m, nothing from USD 14k up clears even 5%.
- At USD 2.5m, the 100 g and USD 20k trades clear 5%.
- The 1 kg sale never clears 5%.

KEY INSIGHTS

- A 100 g sale inside 1% needs a USD 8.08m pool. That is about USD 570 of pool per dollar traded.
- A 1 kg sale inside 5% needs USD 12.1m. That is more than the peak funding Phase 5 measured for the whole business.
- The pool needed grows in step with trade size. There is no saving at scale.

03

The concentrated pool (Uniswap V3)

Capital concentrated around the gold price: the depth it buys, and how often gold moves out of the band.

A V3 pool puts its capital inside a chosen price band.

- AURX should always trade near gold, so the band sits around the gold price.
- Layout: two thirds of the capital in the band, one third across all prices as a backstop.
- Two bands are tested: $\pm 5\%$ and $\pm 10\%$.

Where the liquidity sits

How to read it. Depth at each price, as a multiple of V2, for the same USD 1m. More depth means a trade moves the price less.

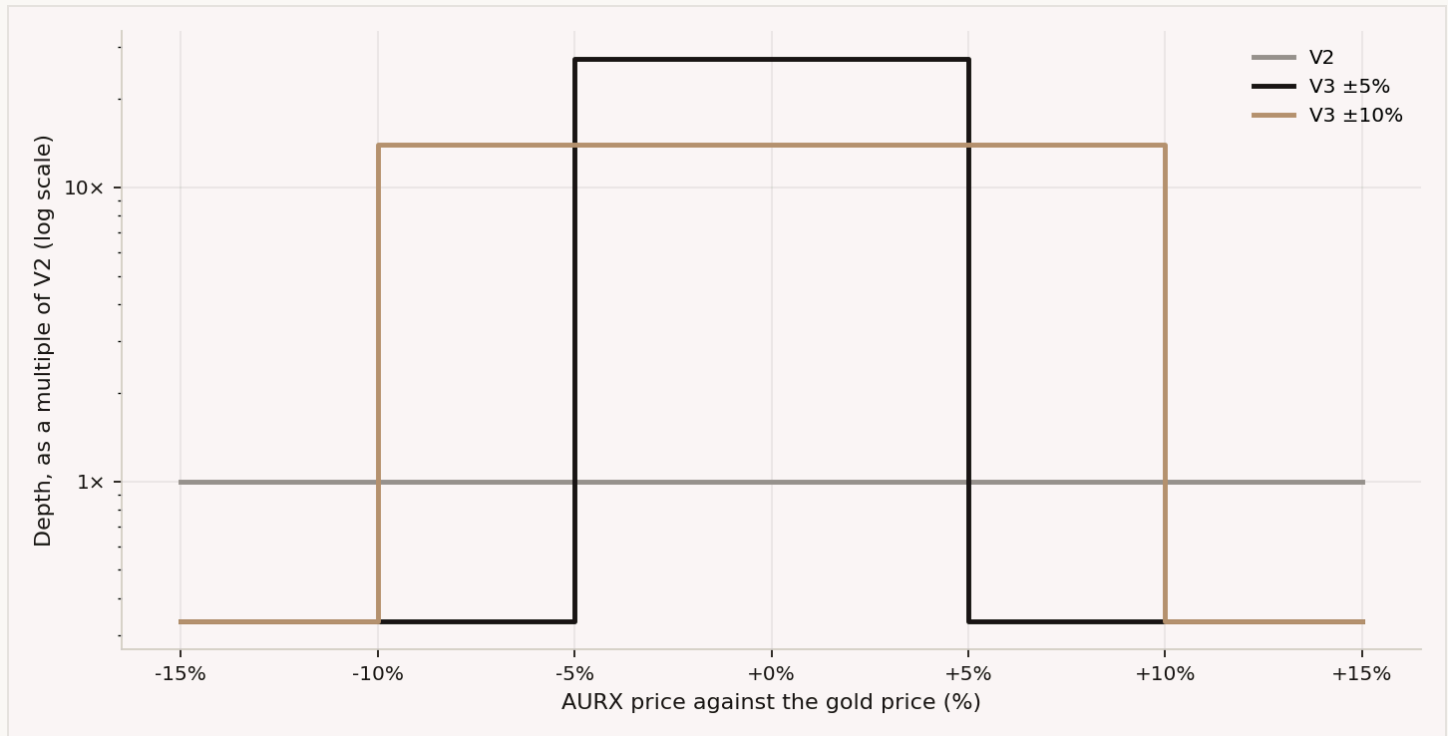


Figure 5. Depth by price, as a multiple of V2, same capital

- Inside the band, $\pm 10\%$ is 14 times as deep as V2.
- Inside the band, $\pm 5\%$ is 27 times as deep as V2.
- Outside the band, both drop to one third of V2. Only the backstop is working.
- A narrower band gives more depth, and less protection when gold moves.

Price impact

How to read it. The same chart as for V2, for the $\pm 10\%$ band.

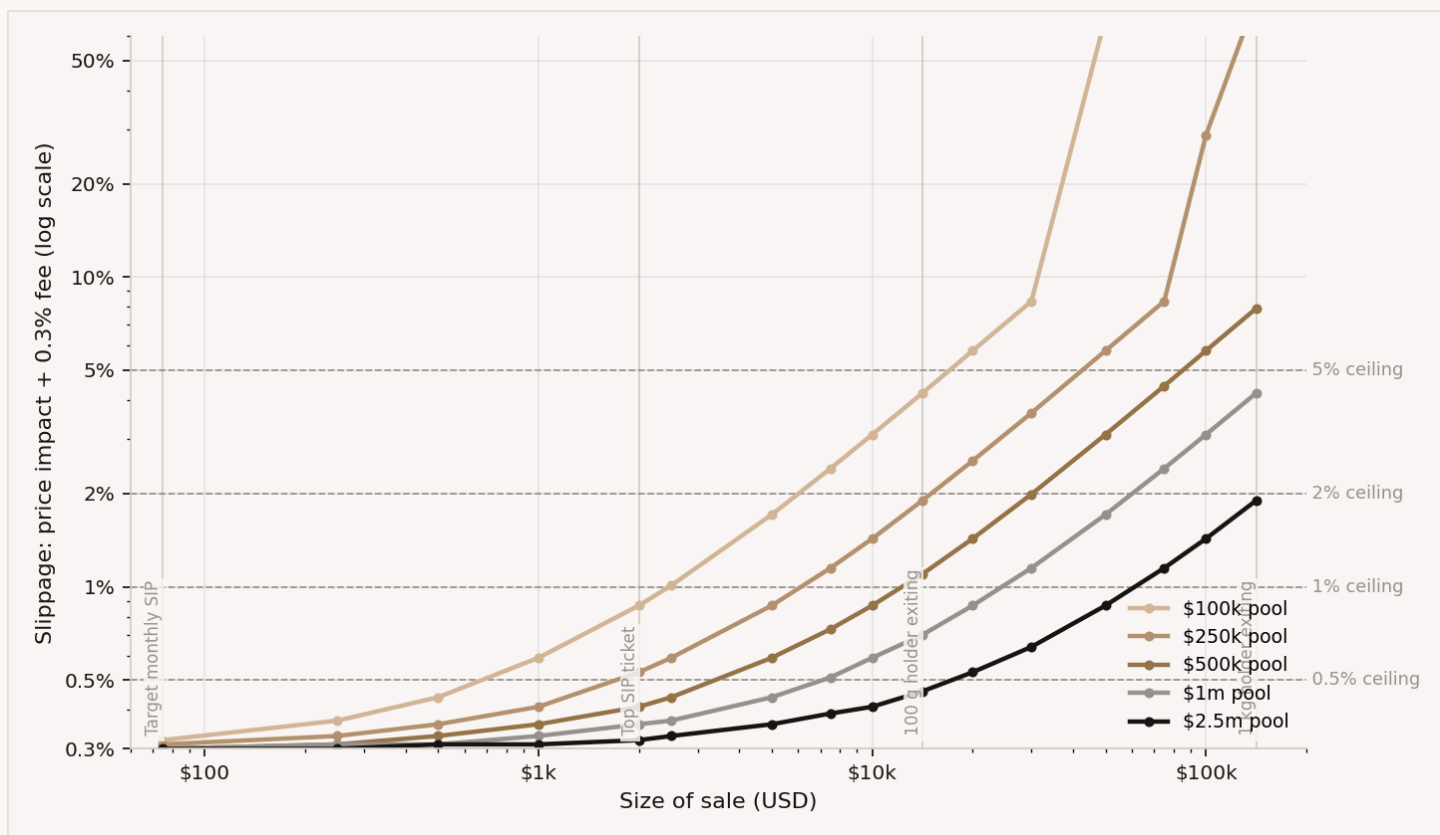


Figure 6. Slippage against sale size in a V3 $\pm 10\%$ pool, one line per pool size

WHAT IT SHOWS (USD 500K POOL)

- USD 2,000 sale: about 0.4%.
- 100 g sale: about 1.1%.
- 1 kg sale: about 7.9%. It needs the USD 1m pool (4.2%) to fit inside 5%.

The two smallest pools bend sharply at the right. There the sale has pushed the price out of the band. That bend is the pool's practical limit.

KEY INSIGHTS

- Everyday trades cost little more than the fee. A USD 2,000 sale in a USD 1m pool costs about 0.36%, of which 0.30% is the fee.
- Costs jump once a trade leaves the band. Size the pool so the largest trade stays inside.

Maximum trade size

How to read it. The largest sale inside each ceiling for a USD 1m pool, V2 against both V3 bands.

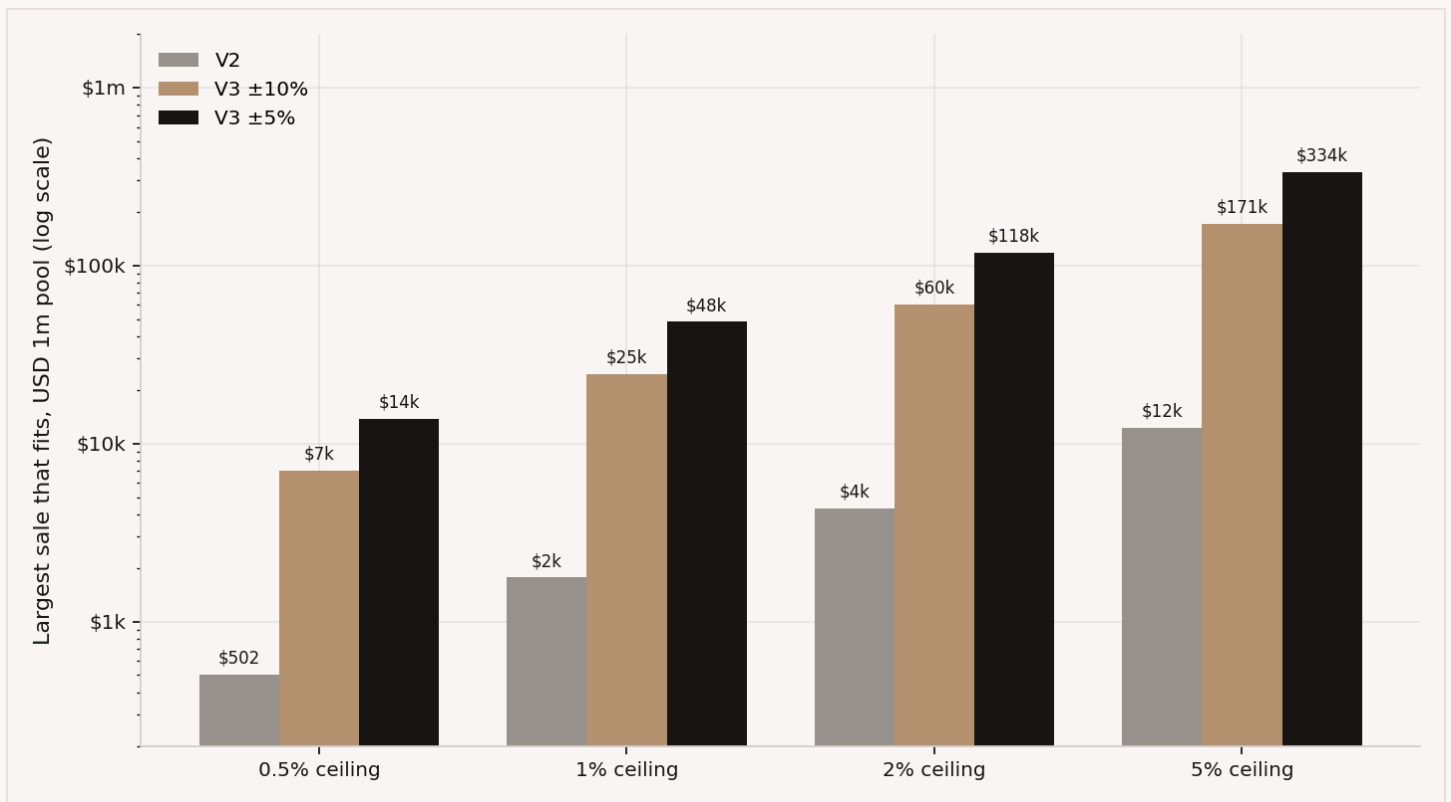


Figure 7. Largest sale inside each ceiling, USD 1m pool, V2 against V3

CEILING	V2	V3 ±10%	V3 ±5%
0.5%	USD 502	USD 7,011	USD 13,718
1%	USD 1,765	USD 24,631	USD 48,194
2%	USD 4,318	USD 60,274	USD 117,934
5%	USD 12,218	USD 170,548	USD 333,700

- ±10% takes 14.0 times more than V2 at every ceiling.
- ±5% takes 27.3 times more.
- At USD 1m, ±10% takes a 1 kg sale inside 5%. V2 cannot do this even at USD 2.5m.

The same grid as in Part 2, for the ±10% band:

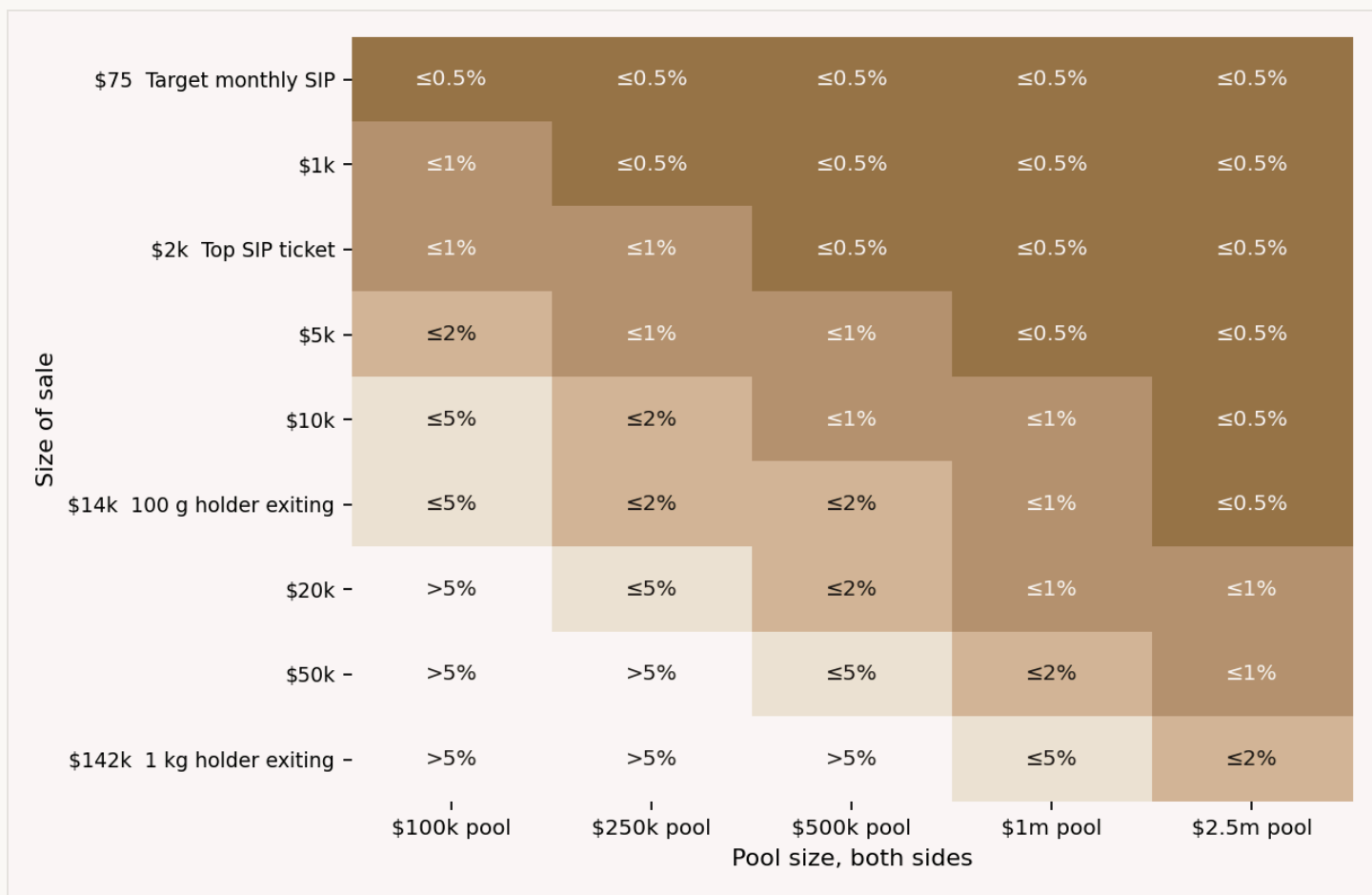


Figure 8. Tightest ceiling each trade clears, by pool size, V3 ±10%

KEY INSIGHTS

- A USD 1m ±10% pool meets the standard benchmark. USD 10k to 50k trades clear inside 2%.
- Halving the band doubles the depth. It only lasts while gold stays inside the band.

How often gold leaves the band

How to read it. Once gold passes the band edge, the band stops working. The chart shows the chance of that within 7, 30 and 90 days (20,000 gold paths at 15% a year).

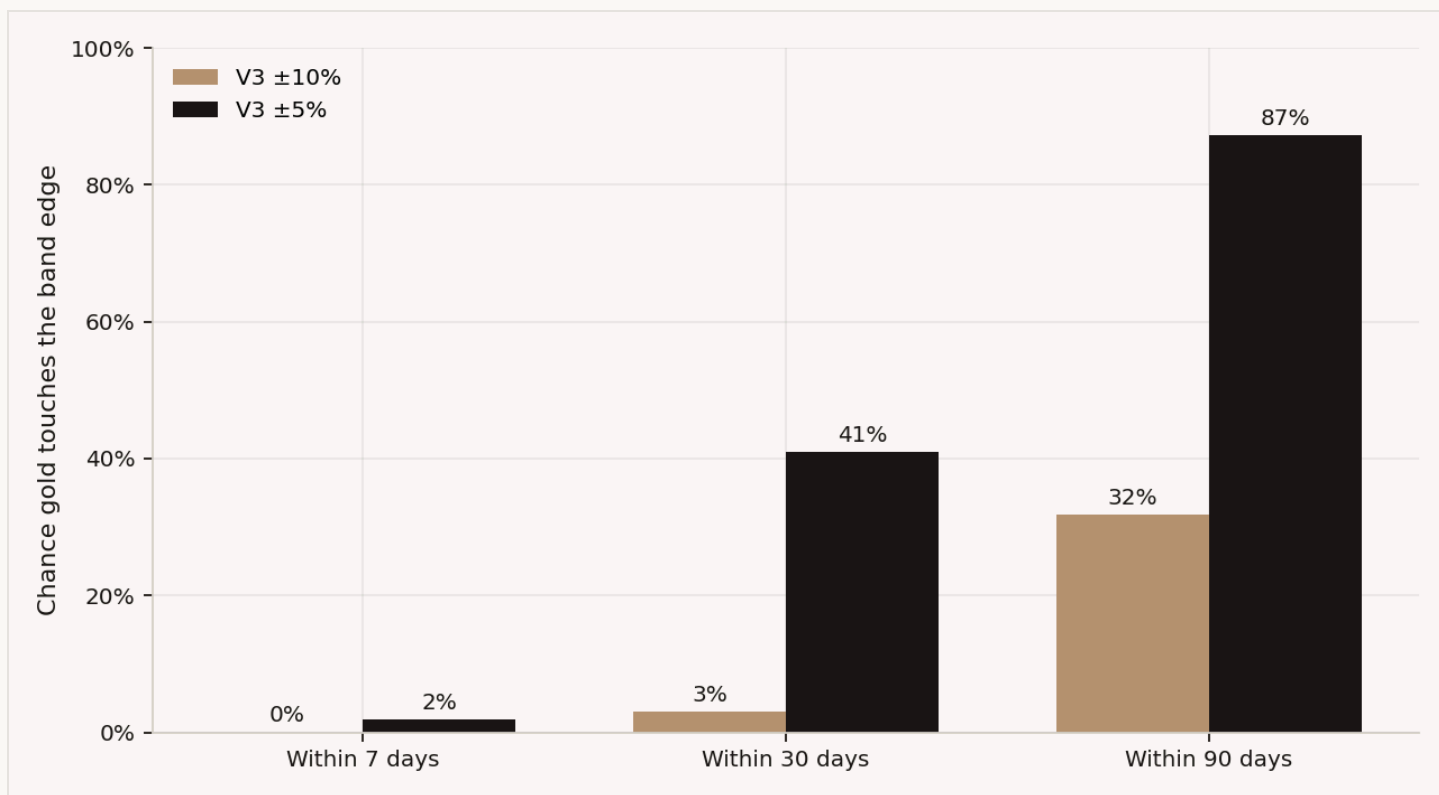


Figure 9. Chance gold leaves each band within 7, 30 and 90 days

BAND	WITHIN 7 DAYS	WITHIN 30 DAYS	WITHIN 90 DAYS
±5%	1.9%	40.9%	87.3%
±10%	0.0%	3.1%	31.8%

After a breach (USD 1m pool, largest sale inside 1%):

BAND	INSIDE THE BAND	GOLD 1 POINT BELOW THE BAND	GOLD 1 POINT ABOVE THE BAND
±5%	USD 48,194	USD 570	USD 606
±10%	USD 24,631	USD 555	USD 620

- **Gold falls through the band:** sellers meet only the backstop. Capacity drops about 85 times (±5%) or 44 times (±10%).
- **Gold rises through the band:** buyers meet the backstop. A larger sale pushes the price back into the band. Inside 2%, a ±5% pool takes USD 55,243 after a rise, against USD 1,395 after a fall.

KEY INSIGHTS

- **±5% needs monthly work.** Gold leaves it within a month in two runs out of five. Someone must re-centre it.
- **±10% needs a quarterly review.** Gold leaves it within a month in 3% of runs.
- Breaches are most likely when holders most want to sell: during a sharp fall in gold.

04

V2 against V3, and the budget

The two designs side by side, the budget for a stated service standard, and what to do.

Side by side

How to read it. All three designs at USD 1m.

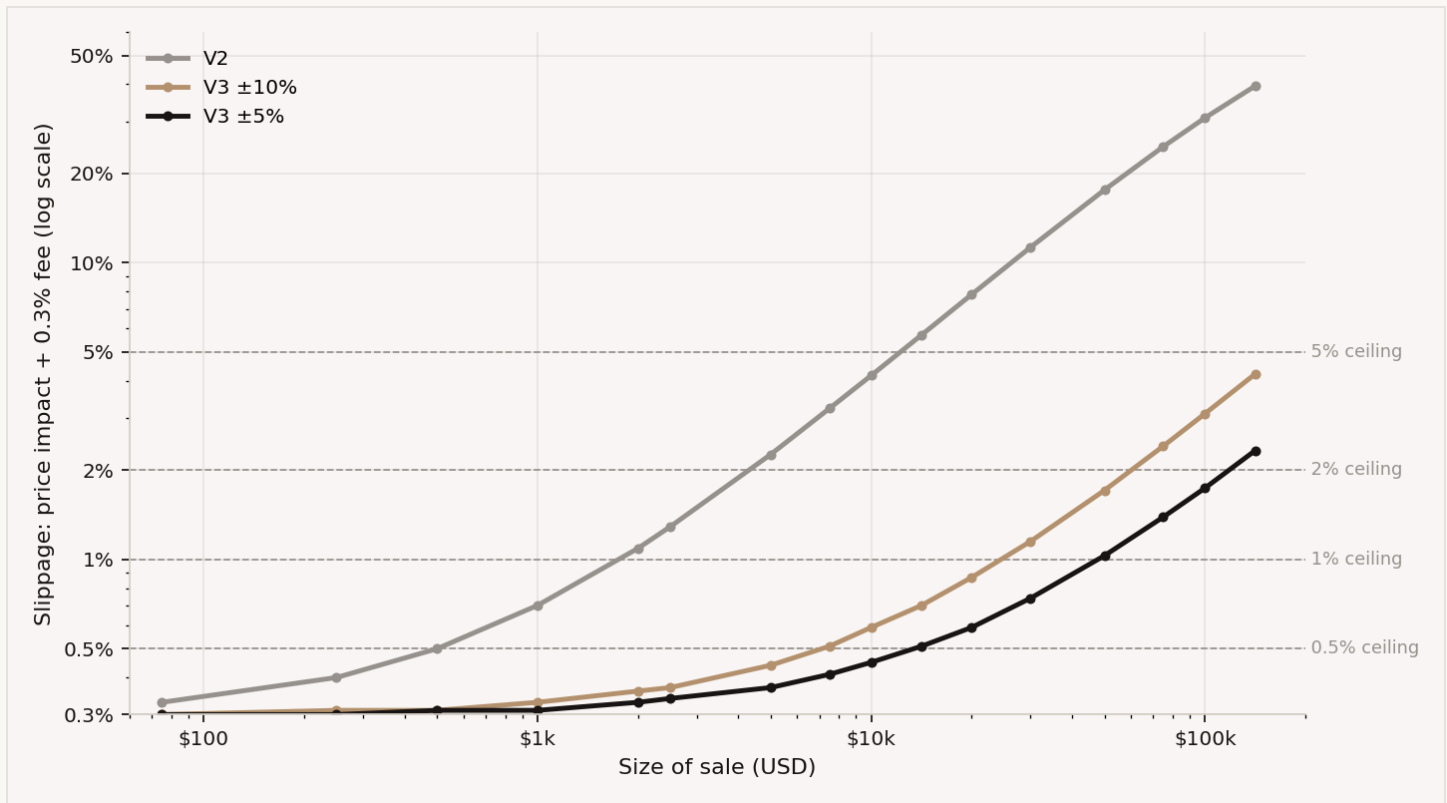


Figure 10. Slippage against sale size, USD 1m pool, three designs

Where each design crosses the 1% ceiling:

- V2: a little above USD 1,700.
- V3 ±10%: near USD 25k.
- V3 ±5%: near USD 48k.

V3 is clearly better. The choice between the two bands depends on how much management Aurumix can take on.

Pool size needed for the ±10% band:

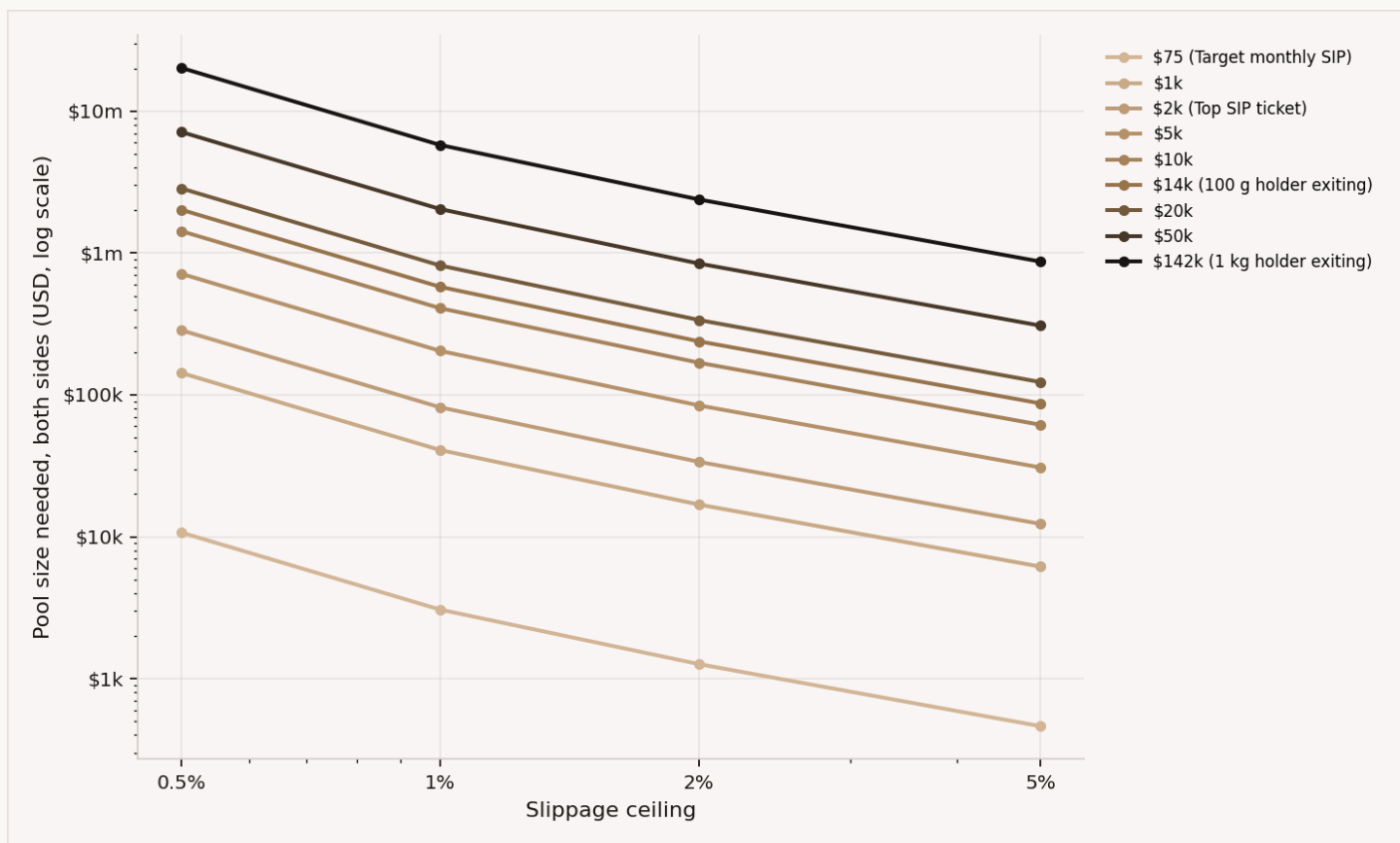


Figure 11. Pool size needed against slippage ceiling, V3 ±10%, one line per trade size

TRADE	0.5%	1%	2%	5%
USD 2,000 (top SIP ticket)	USD 286k	USD 82k	USD 34k	USD 12k
USD 10,000	USD 1.43m	USD 409k	USD 169k	USD 62k
100 g (USD 14,150)	USD 2.02m	USD 579k	USD 239k	USD 87k
1 kg (USD 141,500)	USD 20.2m	USD 5.79m	USD 2.39m	USD 870k

The budget

We set a service standard to size the budget. It is our proposal. If Aurumix picks a different one, the budget scales in a straight line.

LINE	TRADE	INSIDE	WHY
1	Up to USD 2,000	0.5%	Everyday amounts trade at almost the fix
2	100 g (USD 14,150)	1%	A launch-bar holder can exit the same day
3	1 kg (USD 141,500)	5%	The largest bar stays inside the entry fee

The smallest pool that meets all three lines:

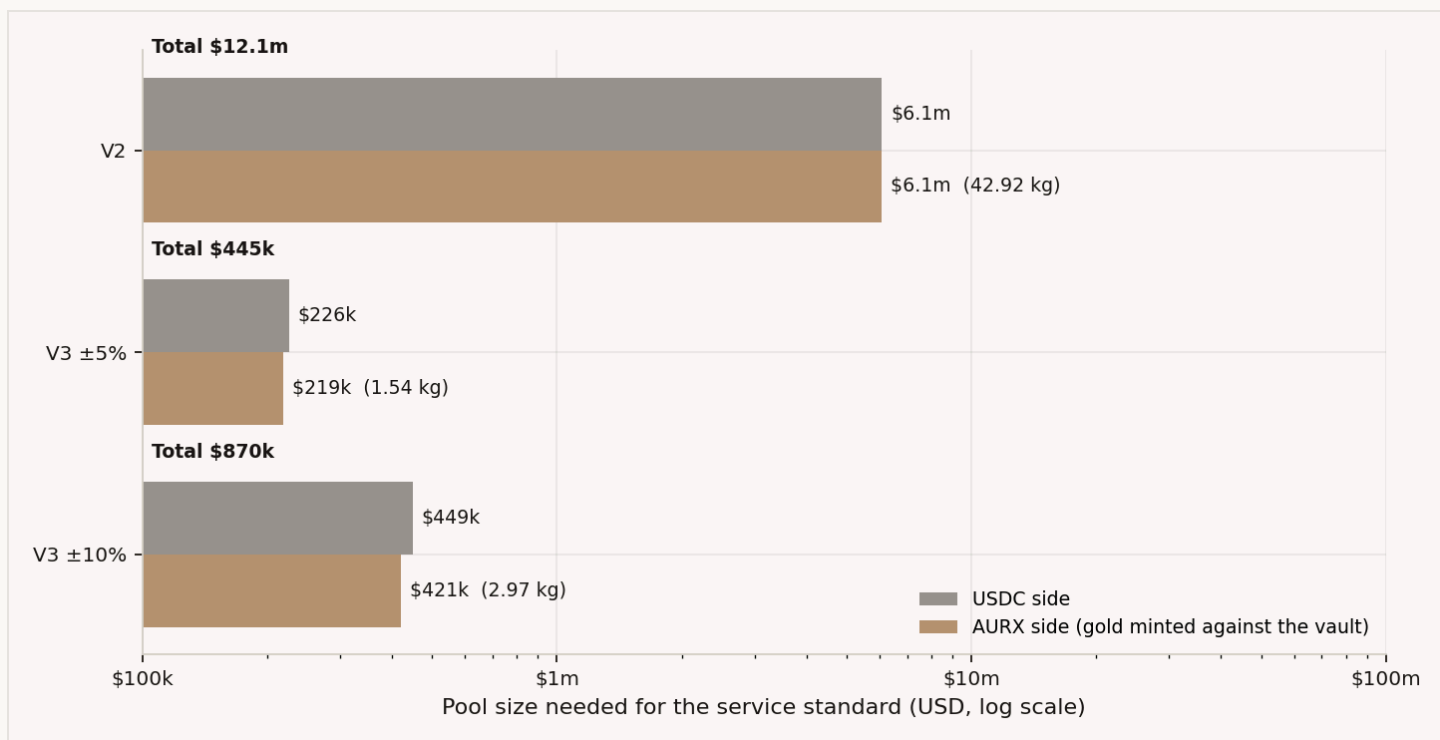


Figure 12. Pool size needed for the service standard, split into USDC and gold

DESIGN	POOL SIZE	STABLECOIN	GOLD	MANAGEMENT
V2	USD 12.1m	USD 6.07m	42.9 kg (USD 6.07m)	None
V3 ±5%	USD 445k	USD 226k	1.55 kg (USD 219k)	Re-centre about monthly
V3 ±10%	USD 870k	USD 449k	2.97 kg (USD 421k)	Review quarterly

- The 1 kg line sets the size in every design.
- Both V3 options give the same service. ±5% uses half the capital and needs monthly work.

What trades cost in the recommended ±10% pool:

TRADE	SLIPPAGE (ENGINE MEASURE)	AVERAGE PRICE AGAINST GOLD
USD 2,000	0.37%	0.33% below
100 g (USD 14,150)	0.76%	0.53% below
1 kg (USD 141,500), sale	4.79%	2.56% below
1 kg (USD 141,500), purchase	5.00%	2.63% above

A stricter standard (1 kg inside 2%) would need:

- USD 2.39m under ±10% (USD 1.23m stablecoin plus 8.16 kg of gold).
- USD 1.22m under ±5%.
- USD 33.3m under V2.

Fee tier

All figures use the 0.30% fee. Uniswap V3 also offers 0.05%.

- A USD 1m $\pm 10\%$ pool at 0.05% takes USD 15,765 inside 0.5% (USD 7,011 at 0.30%).
- Inside 1%: USD 33,408 (USD 24,631 at 0.30%).
- A lower fee earns the pool less. This audit does not measure that.
- The fee tier is a launch decision. It does not change the budget.

Conclusion

Assessment: a pool works as a managed V3 position, once Aurumix can fund both sides.

- A V2 pool would need USD 12.1m. That rules it out.
- A V3 $\pm 10\%$ pool does the job for USD 870k with a quarterly review.
- A V3 $\pm 5\%$ pool does it for USD 445k with monthly work.
- The pool must be managed. It will change how some buyers enter. It opens after the licence and the regulated venues.

STRENGTHS

- Two doors anchor the price: the free buyback below, the app above. The pool only needs depth near one price.
- One fixed budget covers every trade size the product creates. The largest SIP ticket is USD 2,000.
- The gold in the pool is still Aurumix's gold.

RISKS, MOST SERIOUS FIRST

1. **An unmanaged pool shows a discount.** Once gold leaves the band, capacity falls about 44 to 85 times.
2. **The pool undercuts the entry fee.** A buyer who can reach it pays about 0.3 to 0.8% over gold, instead of 5%. The entry fee is the largest retail revenue line.
3. **Breaches come with selling pressure.** A sharp fall in gold triggers selling and pushes the price out of the band.
4. **Aurumix as its own liquidity provider may be a licensed activity.** This is a question for counsel before any pool opens.

Recommendations

Each recommendation is tied to a finding in this audit.

1

Plan for a USD 870k V3 $\pm 10\%$ pool HIGH

- USD 449k of stablecoin plus 2.97 kg of gold, minted as AURX. Review quarterly.
- Move to $\pm 5\%$ (USD 445k) only with a market maker under contract.
- Do not seed a V2 pool.

2

Name the pool's owner before it opens HIGH

- The owner watches gold, re-centres the band, and mints AURX at cost when buyers drain it.
- Write the trigger down. For example: re-centre when gold moves 7% from the band's centre.

3

Open the pool last HIGH

- Order: a VARA-licensed exchange with a market maker, then a broader exchange, then the pool.
- On an exchange, hold about the same value within the same distance of the gold price.
- Seed the pool only when the full budget is available. A half-funded pool is a thin pool.

4

Decide whether the pool may undercut the entry fee MEDIUM

- Near the fix, the pool is cheaper than the app for anyone who can use it.
- Decide this next to the entry-fee calibration.
- Add Aurumix as its own liquidity provider to the counsel list.

05

Appendix

The formulas, the checks against the audit engine, and what this audit leaves out.

Appendix A

Method and formulas

The pool maths reproduces the Tokenomics.net audit engine (`lib/tokenomics_audit/analyses`). The script `liquidity/verify.py` checks every formula against the engine. All checks pass.

Notation. P = price in USD per AURX. f = fee (0.3%). A ceiling s leaves $s - f$ for price movement, as in the engine.

V2 maximum trade (`max_trade_size_analysis.py`). Stablecoin reserve y , AURX reserve x :

- Largest buy: $y \times (\sqrt{1 + s - f} - 1) \div (1 - f)$
- Largest sale, in AURX: $x \times (\sqrt{1 \div (1 - (s - f))} - 1) \div (1 - f)$, valued at P

V2 pool needed (`liquidity_required_analysis.py`): $2 \times (1 - f) \times T \div (\sqrt{1 + s - f} - 1)$ for a buy of T dollars. Where a sale needs more, the larger figure is used.

V3 (`v3_liquidity_analysis.py`). Inside its band, a position with liquidity L behaves as a V2 pool with $L \div \sqrt{P}$ of AURX and $L \times \sqrt{P}$ of stablecoin. Our engine carries a trade across band edges. Two deliberate differences from the engine:

- The engine's V3 module lets price move by the full ceiling s . Its V2 module allows $s - f$. We apply $s - f$ to both, so the comparison is fair.
- Tick spacing is ignored. It moves a band edge by at most 0.6%.

CHECKS RUN

- V2 price moves match the engine to the cent at 0.5, 2 and 5%.
- V2 maximum trades and pool sizes match at all four ceilings.
- A V3 trade inside its band matches a V2 pool on the band's reserves.
- Capacity triples when capital triples.
- The V3 layout hits its pool size and its two-thirds split.

Band breach. 20,000 gold paths, daily steps, 15% a year. A breach counts when gold touches the band edge on any day.

Reproduce. In `deliverables/3-supply-side-tokenomics/liquidity/`, run `python verify.py`, `python run.py`, then `python charts.py`. All inputs are in `params.py` with sources.

Appendix B

What this audit does not cover

- **Allocations, vesting, discounts, returns, unlocks.** AURX has none.
- **The liquidity provider's profit and loss.** Fee income and losses from gold moves are not measured.
- **Re-centring cost.** Each reset trades part of the position. Not measured.
- **Order-book exchanges.** They use market makers. The depth figures are a rough guide.
- **Chain and gas costs.** No chain is assumed.
- **Legal questions.** Aurumix as its own liquidity provider, and whether a listing changes how AURX is classified, are for counsel.

tokenomics.net

Built by founders. Designed for founders.